

Nirmal Francis Xavier

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EDUCATION

Georgia Institute of Technology | MS Computer Science Dec 2026

GPA: 4.0/4.0 | ML, Deep Learning, Design of Algorithms, Data & Visual Analytics, Knowledge based AI, Big Data Systems

Indian Institute of Technology Kharagpur | MTech Industrial Engineering May 2019

GPA: 9.57/10 | Operations Research, Statistical Decision Modelling, Systems Modelling and Analysis, Safety Analytics

SKILLS

Programming & Tools: Python, R, SQL, PowerBI, Git, Excel, C++, Pyspark, Javascript, AWS, Docker, DataBricks

Core Competencies: Machine Learning, Optimization, Markov Chain Modelling, Monte Carlo Simulation, Hypothesis Testing, Data Engineering, Deep Learning

EXPERIENCE

RELEX SOLUTIONS | Intern - Supply Chain Analytics May 2026 - Aug 2026

- Implemented a distribution-center-level pooled Bayesian regression forecast for a North American auto parts retailer; evaluated against traditional time series models, improving WAPE by 5% and reducing over-bias by 8%.
- Investigated inventory and order-proposal data to identify and quantify leading causes of spoilage and stockouts for a Central American membership-based warehouse club.

PROTIUM FINANCE | AVP - Data Science Dec 2022 - Jul 2025

- Built XGBoost binary classifier on AWS SageMaker to predict 18-month repayment risk; reduced 8,000+ features to 13 using Optuna + feature selection (F-score, IV, SHAP), achieving a test AUC of 0.698; used the score to drive lending policy.
- Developed a monthly repayment risk score using Logistic Regression (test ROC AUC: 0.876), with WOE-binned features for robustness and VIF-based feature reduction to minimize multicollinearity; operationalized the score in a prioritization strategy that cut operating costs by 15% and improved on-time repayment outcomes by 5%.
- Built Markov-chain loss forecasting model using roll-rate transitions to project lifetime NPA by segment; used the forecasts to flag high-risk cohorts and guide policy tightening, reducing overall NPA by ~12%.
- Implemented rule-based fuzzy matching algorithm combining SBERT + FAISS retrieval with string-similarity metrics (WRatio, token set, Jaro Winkler) to automate customer verification, reducing manual underwriting by 3000 hours annually.
- Owned the Portfolio Power BI dashboard with 100+ measures, centralizing portfolio, collections, and productivity insights.
- Designed a repayment-score driven collections allocation strategy, cutting ops cost by 15% and flow forward rates by 5%.

CUSHMAN & WAKEFIELD | Senior Consultant - Data Analytics Jan 2022 - Dec 2022

- Built an end-to-end maintenance staffing optimizer for HDB (Singapore): facility-location model to place standby units, discrete event simulation (SimPy) for optimal no. of technicians by site and shift from demand and service times, and a roster optimization model (PuLP) with labor + coverage constraints to cut operational costs by \$1M/year.

GENERAL MILLS | Senior Engineer - Manufacturing Analytics Oct 2019 - Jan 2022

- Developed real-time defect prediction model using Random Forest for rare “cheese oiling-out” failures using time-lagged process signals. Model optimized for failure capture with F0.5 and delivered ~\$1M annual savings by preventing failures.
- Built mixed-integer programming model in R to optimize changeover sequencing and assign product flow across processing, surge capacity, and packaging lines. Minimized runout waste and achieved ~\$500k/year savings.

SELECTED PROJECTS

- Image Captioning Evaluation | Compared CNN-LSTM, attention LSTM, and Transformer models on MS COCO using Karpathy splits and four training-data fractions; evaluated captions with BLEU, METEOR, CIDEr, and ROUGE-L and inspected training curves and attention visualizations.
- Built product substitute recommendation system using Gaussian Mixture Model to learn soft product families and generate high-quality substitutes. Best project award, Georgia Tech CS7641, Fall 2025

CERTIFICATION (Links)

SQL (Advanced), Hacker Rank, Feb 2024 | Supply Chain Digitization and Management, IIM Mumbai, Mar 2021